



Comparing a Human's and a Multi-Agent System's Thematic Analysis: Assessing Qualitative Coding Consistency

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Abstract. Large Language Models (LLMs) have demonstrated fluency in text generation and reasoning tasks. Consequently, the field has probed the ability of LLMs to automate qualitative analysis, including inductive thematic analysis (iTA), previously achieved through human reasoning only. Studies using LLMs for iTA have yielded mixed results so far. LLMs have successfully been used for isolated steps of iTA in hybrid setups. With recent advances in multi-agent systems (MAS) enabling complex reasoning and task execution through multiple, collaborating LLM agents, the first results point towards the possibility of automating sequences of the iTA process. However, previous work especially lacks methodological standards for assessing the reliability and validity of LLM-derived iTA. Thus, in this paper, we propose a method for assessing the quality of iTA systems based on consistency with human coding on a benchmark dataset. We present criteria for benchmark datasets and an expert blind review with this method on two iTA outputs: one iTA conducted by domain experts, and another fully automated with a MAS built on the Claude 3.5 Sonnet LLM. Results indicate a high level of consistency and contribute evidence that complex qualitative analysis methods common in AIED research can be carried out by MAS.

Keywords: Large Language Models · Qualitative Coding · Thematic Analysis · Inductive Analysis · Claude 3.5 Sonnet · Agentic LLMs · Multi-Agent Systems

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1 Introduction

Educational researchers often analyze qualitative data, such as open-ended surveys, interviews, observation notes, classroom interactions, and written assignments. In the field of Artificial Intelligence in Education (AIED), these data are often used to understand learning, engagement, self-regulation, collaboration, and other key processes occurring in and around AIED learning environments [1–3]. Qualitative data analysis, whether used on its own or mixed with quantitative data analysis, offers deeper insights into the learning process. However, the volume and complexity of such qualitative data pose substantial challenges for manual analysis (methods scale poorly with dataset size and ideally involve multiple researchers to reduce bias), making automation an attractive prospect.

The rapid advancement of large language models (LLMs), particularly for natural language processing and structured text generation, has created the possibility of automating qualitative analysis. Having been trained on massive amounts of text data, LLMs have shown the ability to recognize patterns in data, even beyond their training sets [4]. This makes them promising candidates for assisting or even fully automating interpretative research tasks.

However, a hallmark of qualitative research is not just its “correctness” [5], but rather its depth, nuance, and the ability to tell “stories” about the data [6]. Consequently, researchers have been investigating whether LLMs can reliably extract meaningful insights from qualitative data while maintaining the depth and nuance typically associated with human analysis [7]. So far, these studies have typically employed LLMs for isolated steps in hybrid procedures (assisting human coders) [8] and for well-defined artifacts, such as labeling data with codebooks [7]. Therefore, to advance the state-of-the-art, the first challenge is to build a fully automated AI system that may take over a larger part of the full process without human intervention. The value of such a system is only realized if such a system maintains coherence, a level of depth, and nuance in the AI-generated codes comparable to expert-generated ones. However, the criteria for a comparison along these dimensions are lacking. Finally, such a comparison requires a reference dataset with a sufficient level of depth and nuance to enable a meaningful comparison. Addressing these gaps constitutes the core contribution of this paper.

As an example of a complex, yet common qualitative method in AIED, requiring dynamic discussion between human coders of qualitative data to derive research findings and implications, we chose inductive Thematic Analysis (iTAT). TA is an iterative six-phase process involving familiarization with data, the generation of codes, the construction (definition and naming) of themes, as well as the definition and naming of themes [9, 10]. TA can be grounded in two approaches: inductive or deductive. Inductive (sometimes called “indicative” in the literature) analysis involves a bottom-up approach, where codes and themes emerge directly from the data without an explicit, a priori theoretical framework. In contrast, deductive analysis follows a top-down approach, where the analysis is driven by pre-existing theories or frameworks, and themes are aligned with those predefined theoretical constructs [9]. Irrespective of which approach is chosen, TA traditionally requires significant manual effort, typically involving collaboration between at least two researchers to enhance reliability and achieve consensus through iterative coding [11].

To replicate and automate the iTA process, we use a recently reported multi-agent LLM workflow [12]. Given a complex task, a multi-agent approach involves an “orchestrator” agent that breaks down the task into subtasks to be executed by agents playing different roles, each specialized in accomplishing different subtasks [23, 24]. The agentic workflow can, therefore, spin off specialized agents for various aspects of iTA, such as coding, consensus-building, and identification and review of themes. By modeling human consensus-making processes as closely as possible, our agentic workflow has a fairer opportunity to achieve analytical nuance rivaling human coding than workflows with single prompts or unidirectional processing of data segments [12].

To evaluate the performance of this fully automated workflow, that is, to determine the degree to which automatically derived themes and codes align with results of expert human coders, we run the workflow on a survey dataset of responses from researchers in the field of Computer-Supported Collaborative Learning (CSCL) [13]. As a domain-specific dataset requiring domain knowledge for adequate and nuanced interpretation, the dataset serves as an appropriate testbed for the comparison. We further tasked two independent experts to assess both outputs blindly through templates integrating the developed criteria, the results of which we report in this paper alongside the experts’ feedback on the method itself. These contributions stem from the following research questions:

RQ1: What constitutes an appropriate *method* to assess the quality of iTA?

RQ2: To what extent can the inductive thematic analysis process be automated?

2 Related Work

The advent of LLMs, including open-source options like Llama and API-accessible closed-source options like ChatGPT, Mistral, and Claude, has accelerated the traditional time-consuming process of both inductive and deductive thematic analysis. Several studies [7, 8] have harnessed the potential of LLMs for performing TA-related tasks. For example, Barany et al. investigated the application of GPT-4 for inductive codebook development, focusing on its ability to address pervasive challenges in manual code generation, such as time constraints, inconsistencies, and potential human error during the coding process [7]. Their results emphasized that while the hybrid approach (i.e., collaboration between humans and AI) strikes a balance between efficiency and reliability, fully automated approaches (i.e., solely relying on GPT), although time efficient, have limitations in terms of clarity, mutual exclusivity, and reliability. Similarly, Paoli not only employed GPT-3.5 Turbo to conduct an iTA on two datasets of semi-structured interviews, 13 interviews with video game players and 10 interviews with lecturers teaching data science, but also detected which phases of the TA process are challenging for LLMs [14]. Paoli emphasized that phase one (i.e., familiarizing with data) and phase six (i.e., generating the report) require human involvement.

On the other hand, several studies have explored the use of LLMs for deductive qualitative analysis [15, 16]. For example, Xiao et al. adopted GPT-3 for deductive coding tasks, reporting agreement with human coders on question complexity (Cohen’s $\kappa = 0.61$) and fair agreement on syntactic structure (Cohen’s $\kappa = 0.38$) [16].

In sum, most related studies explore various versions of GPT and their potential to assist in developing codebooks for both inductive and deductive TA, often evaluating

and comparing the generated outcomes with those produced by human analysts to assess reliability, validity, and interpretative depth. In this regard, Yan et al. emphasize that GPT falls short in several critical areas, including trustworthiness, reliability, consistency, data capacity, contextual understanding, and interface limitations [17]. Supporting these findings, Tai et al. explored the use of GPT-3.5 for deductive coding and reported that, although its coding performance was comparable to traditional human coding, significant limitations persisted, such as algorithmic constraints and token limits (i.e., GPT-3.5 can generate a maximum of 2048 characters per input) [18]. Hence, the methods in these prior studies are neither fully automated nor have they produced consistently good results.

3 Method

3.1 Dataset

A dataset to benchmark iTA systems should be difficult to analyze, meaning that it requires substantial interpretation from domain experts to derive insights from the data. This ensures that the performance of LLMs on the task is based on reasoning and analysis as opposed to the recall of already-available information in the training data used in the LLMs. We contribute a novel set of criteria for benchmarking datasets to ensure this property is met:

1. **Insight Novelty:** The insights generated from the data are not public knowledge or known a priori, meaning the findings can only be obtained from the specific data provided and not through prior knowledge alone.
2. **Domain Specificity:** If the data are domain-specific, expert knowledge is required for accurate coding and extracting meaningful themes. Data meeting this criterion demands knowledge beyond general understanding and thus serves as a higher bar for “general purpose” LLMs to clear.
3. **Semantic Proximity:** If data expose subtle differences between concepts or instances, analysis must carefully differentiate these nuances and maintain them throughout the iTA process to tell equally nuanced “stories” of what the data represent [6]. This criterion applies when distinct items possess very similar meanings, requiring close examination during the iTA process.
4. **Data Point Length:** The amount of information provided in a data point impacts analysis; more information generally provides better context for identifying codes, while shorter data points tend to leave more room for different interpretations.
5. **Conceptual Novelty:** If concepts in the data are novel, emergent, or not widely agreed upon, coders must consider different possible meanings. This criterion signifies that the data contains ideas requiring definition or interpretation during analysis.

In addition to defining these criteria, we share one such dataset that meets the criteria: a survey of CSCL scholars collected in prior work [13]. The survey investigated prevailing definitions among the CSCL community for CSCL itself (collaborative learning, the differences between collaborative and cooperative learning, and researchers’ perspectives on the future of the field). Both the manual and automated TA were conducted on a total of 184 responses to four open-ended questions in this survey in Fall 2024. The

survey featured consent to full anonymous publication and reuse of the survey data. The dataset is available on the [OSF](https://osf.io/fbw3n/) platform.

The dataset meets the criteria described above: definitions required domain experts for the manual iTA to distinguish between the subtle semantic differences, and the definitions were both short and perceived as nuanced by the iTA team conducting the manual analysis. The survey was conducted based on the hypothesis that consensus for these concepts was absent in the CSCL community, covering both the criterion for missing conceptual stability and insight novelty.

3.2 Manual iTA Method

The manual iTA results used in this study were originally published by Simon et al. [13]. Their manual iTA was conducted along the six-step schema of Braun & Clarke to identify common or conflicting themes, categories, and codes in qualitative data [9]. To analyze 184 open-ended survey responses (excluding 10 for minimal content of a few or no words), a team of four CSCL experts engaged in individual open coding [9]. According to the results of Simon et al.’s study, three coders independently reviewed each survey response, identifying meaningful themes and assigning codes in separate documents. Subsequently, the three coders, along with a fourth researcher, collaboratively merged similar codes and resolved interpretive discrepancies in a shared spreadsheet to arrive at the final, refined coding framework. Each researcher then used the refined coding framework to identify themes within each survey question independently. Finally, all coders discussed and established themes and descriptions from the codes. The intercoder reliability was 0.934 (Krippendorff’s α) [13, 19, 20].

3.3 Multi-Agent System Architecture for iTA

The tested system is a multi-agent system (MAS), built around Amazon Bedrock Agent’s custom orchestrator workflow¹ with individual LLM calls processed by the Claude 3.5 Sonnet model. The model was selected for its reasoning performance on text (DROP, F1 Score: 87%) compared to other models like Open AI’s ChatGPT 4o (81%) [26]. MASs consist of multiple autonomous agents interacting and communicating to achieve individual or collective goals, coordinated through an orchestration agent [21].

The system’s input is a detailed description of the iTA process, including all intermediate steps. This is considered the high-level “agentic” prompt. In the first LLM call, the orchestrator agent parses this description into individual tasks that are then assigned to independent agents. The number of agents “spawned” by the orchestrator agent depends on the process and description. In our case, we described the Braun & Clarke process that was followed in the Manual iTA Method section above², alongside a description of the dataset (CSCL definitions as free-form answers to questions). Using the process description, the orchestrator agent spawned three individual “coding” agents and a fourth, “consensus” agent, to merge similar codes and resolve interpretive discrepancies

¹ <https://aws.amazon.com/blogs/machine-learning/getting-started-with-amazon-bedrock-agents-custom-orchestrator/>.

² Available at <https://osf.io/fbw3n/files/osfstorage/6813bac5b05dc3cc159fde58>.

(Fig. 1). The process is, thus, analogous to human coding during TA; human coders process information independently before sharing it to establish consensus on codes. This multi-stage process of coding and sensemaking, with individual and collective stages, is reflected by the MAS architecture as well. The system begins with a high-level “agentic” prompt to decompose the iTA process into atomic tasks, which are then executed independently by different agents (Fig. 1).

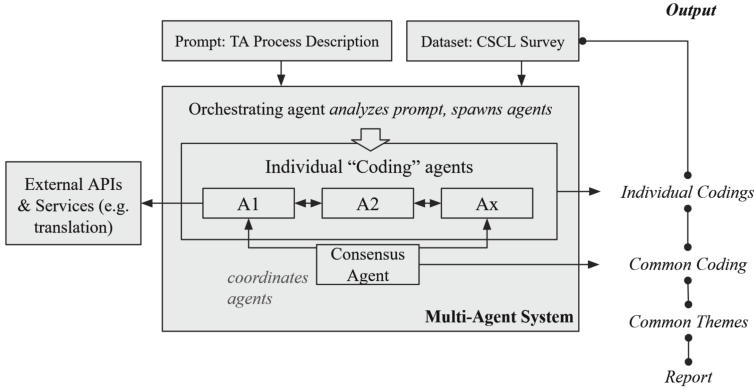


Fig. 1. Architecture of the automated agent-based system for iTA

These tasks include identifying meaningful text units, translating any that might be in a language other than the rest of the data, tagging them with codes, and organizing these codes into themes in a bottom-up fashion. Once the orchestrator agent has taken the “agentic” prompt and decomposed it into the tasks involved in the iTA process, each task is performed as a separate LLM call by a specialized agent. Some steps wait for the outputs of previous steps to complete, since they use these intermediate outputs as their inputs. As an example, the consensus agent waits for the outputs of the three individual coding agents before initiating the consensus-building and reconciliation LLM call. The performance improvement from the use of “agentic” prompts and the multi-agent system architecture is thought to be from a mechanism similar to chain-of-thought prompting. Chain-of-thought prompting consists of prompting LLMs to generate intermediate steps while problem-solving for better overall performance [22]. Similarly, the MAS approach breaks down the analysis into smaller, more manageable parts, reported as leading to better reasoning and increased clarity in prior work [23, 24]. Importantly, the system can dynamically adjust the order of these tasks depending on the complexity of the data, creating a flexible, graph-like structure instead of a rigid, linear one. For instance, a portion of the responses were in French, so the MAS system automatically added a “translator” agent to translate those responses to English before the rest of the process continued. This represents the possibility of fully automating otherwise tedious steps in the process, although we note the limitation of outputs needing to be manually verified to reach confidence in the veracity and real-world range of such a system.

3.4 Method for Comparison with Manual iTA

Both human iTA and automated systems produce reports including themes, which in turn are composed of the codes a theme represents, a description, and a name. Every data point is represented through one or multiple codes.

We reached out to two external experts to rate the output of MAS and human iTAs following the method described in the subsequent paragraphs. Both experts hold expertise in the domain of CSCL and more than five years of experience in the field. Expert input was collected independently of the data and theme curation through the MAS. We asked the experts to evaluate both the codes and the themes through a blinded template containing data from automated and manual TA and criteria for evaluation, without disclosing which codes and themes were derived by which method. The order in which themes were presented was randomized. The raters were not given a codebook. Experts were presented with the code part first to familiarize themselves with the underlying dataset, to later enable the assessment of theme representativeness.

Evaluation of Codes. For the evaluation of qualitative codes, experts had to rate individual code validity, assessing whether a set of codes correctly reflected the data point or not, and if not, justify. In a second step, comparative code quality was evaluated: experts indicated if one of the two proposed sets of codes reflected the associated survey response better, or if both reflected the content equally well (or if neither did). By establishing this baseline, along the provided justifications, perceived differences in the direct comparison can be put into perspective (major or minor). We further assessed bias (e.g., an expert being aware whether code sets were created by AI or humans, albeit randomization of the two columns in the template) by asking experts to rate whether they thought the codes were AI-generated. The following is an example of a code assessment for the question “How do you define Computer Supported Collaborative Learning (CSCL) in one sentence?” (bold: expert response).

- Survey Response: “Leveraging technological tools and practices to facilitate/ support collaborative learning processes and outcomes”.
- iTA codes 1: “C1: Technological tools to support CL process, C2: Technological tools to support CL outcomes, C3: Technological practices to support CL process, C4: Technological practices to support CL outcomes”.
- Do you think the codes of Thematic Analysis 1 (Column B) reflect the response (Column A)? Agree: 1 Disagree: 0: “**1**”.
- If not, please write the reason for your disagreement: “ ”
- “iTA codes 2”: C1: Digital tools for learning, C2: Collaborative processes.
- Do you think the codes of Thematic Analysis 2 (Column B) reflect the response (Column A)? Agree: 1 Disagree: 0: “**1**”.
- If not, please write the reason for your disagreement: “ ”
- Which set of codes better reflects the data point? TA1: 1 TA2: 2 Both do: 3 None does: 4): “**2**”.
- Please write the reason for your choice (besides 3): “This option captures the difference between process and outcomes.”
- Which of the codes do you think were created by AI? TA1: 1 TA2: “**2**”.

Evaluation of Themes. While code sets were presented (shuffled) side by side for each data point, the template for the theme assessment presented all themes, their descriptions, and associated code sets in a single, shuffled list. This design decision mirrored the fact that different iTAs can produce a different number of themes (and any alignment from our side, similar to the side-by-side code set presentation, would have constituted an influence on the experts). Experts instead were asked to identify similar themes (and provide the number of the themes they considered related). Thereby, we could check for possible alignment of themes between AI and human iTA. Experts were asked to rate the identified alignment (from 1 – “somewhat related” to 5 – “equivalent”).

We further asked experts to rate each theme along the dimensions of clarity (theme description), representativeness (alignment of theme description and name with its codes), and relevance (whether the theme covered an important aspect of the dataset). Again, experts were asked if the themes were thought to be AI-generated or not.

The criteria for evaluation of themes were informed by literature assessing the quality of iTA [7] and, given the variety of approaches in existence, through exchanges with experts in the field. The following is an example of a theme assessment (bold: expert response):

- Theme Name: “Group Focus”
- Theme Description: “CSCL acts on different dimensions of groups and collectives, not limited to learning but also work.”
- Codes: “Group size, Community, Group-based learning, Group work”
- Clarity: Is the theme description easy to understand?: “**yes**”
- Representativeness: Does the theme align with the associated codes?: “**yes**”
- Relevance: Does the theme capture an important aspect of the dataset?: “**yes**”
- Is there another theme in this list you think is expressing the same theme? Specify its number: “”
- If yes, to what degree do you think these themes are similar (on a scale from 1 - somewhat related to 5 - equivalent): “”
- Do you think this theme was created by AI?: “**yes**”

4 Results

4.1 RQ1: A Method to Assess the Quality of iTA

In the previous section, we presented the method we applied in this paper to assess the output of an automated system tasked to execute an iTA, based on a manual, blindfolded, standardized process. This process was supported by a template containing output data from both iTAs and questions to guide experts in the assessment of code and theme quality, both intra- and inter-analysis-wise.

The criteria for assessing quality, distilled from literature and expert input described in the previous section - individual code quality assessment, double-blind AI vs. manual code comparison, and theme evaluation based on representativeness, clarity, and relevance—allowed to apply the method on a benchmark dataset, introduced in the method section of this paper. The chosen dataset satisfies desirable aspects, notably insight novelty, data domain, semantic distance, data point length, and conceptual stability.

4.2 RQ2: Comparison Between Manual and Automated iTAs

Codes. The automated system produced an average of 1.7 codes per data point, compared to an average of 2.5 codes by the human iTA team. Both humans and the system produced the most codes (AI 2.5, human coder team 3.3) for question 2 (definitions for collaborative learning). Coder one incorrectly identified AI and human-generated codes for 54% of all data points. Coder two incorrectly attributed code sets in 84% of all cases (85%, 94%, 76%, 81% for questions one to four), systematically picking the wrong option. This significant bias was confirmed through an exact binomial test with $p < 0.05$, $n = 184$.

Tasked to assess if the individual code sets correctly reflected the data, experts positively rated human codes on average 81%, whereas AI codes were considered appropriate in 91% of all data points. Except for human codes for question three, coder two rated more codes as appropriate than coder one, with an average difference of 8% (Table 1). When experts compared both AI and human code sets, coder two considered both sets as equally representative in 90%, 92%, and 88% for questions one, two, and four, respectively. Coder one instead preferred human codes for Q1, Q2, and Q4 (87%, 71%, 56%). For question three, coder one and coder two preferred AI codes for 68% and 50% of all replies (Fig. 2).

Table 1. Experts’ assessments of whether AI or manual codes better represent the data points. “both” and “neither” were alternative options.

Coder	Question	AI Better	Both Good	Humans Better	Both Bad
Coder 1	Q1	2%	6%	87%	4%
	Q2	15%	10%	71%	4%
	Q3	68%	11%	14%	8%
	Q4	13%	27%	56%	4%
Coder 2	Q1	4%	90%	6%	0%
	Q2	0%	92%	8%	0%
	Q3	50%	42%	8%	0%
	Q4	13%	88%	0%	0%

Coder one provided for 33% of data points the explanation that the manual code sets better caught the “nuances” in the data. For question two, a recurrent (39%) comment was that the codes better captured the entirety of the data point. For the preference of AI codes in question three, coder one mentioned that the codes better captured the relevant part of the data points.

Combining both experts’ assessments, a preference for human or both codes for questions one, two, and four seems to prevail, whereas question 3 has more codes better represented by AI. While this seems to contradict the first analysis, where more AI codes were considered to reflect the data points, it shows the distinction between code coverage and relative code set quality.

Themes: Generally, the automated TA system produced fewer themes than the human coder team (on average 5 and 9.5, respectively). On average, experts identified themes correctly in 44% (both experts) as AI or human creations. However, there are stark contrasts between questions. Coder one, for instance, correctly rated 4% of themes for question three, and coder two rated correctly 91% of themes for the same question, 8% for question two, and 17% for question four. A binomial test confirmed these cases with $p < 0.05$. While coder two assessed both AI and human themes positively for clarity (100% of themes rated positively), coder two, on average, rated more AI themes positively (90%) compared to human themes (73%). The difference, though present (95% for AI themes against 91% for human themes), was less for the number of themes assessed for relevance. For this criterion, coder two did not rate AI and human themes differently (80% for both human and AI themes). Similarly, no difference was present for AI and human-coded themes for coder two's rating of representativeness. Coder one only found 36% of themes representative of question two, but without a difference in the global average.

When experts were asked to identify which themes were related, the number of combinations was 22 (18 by coder one and seven by coder two). Experts identified three of these combinations conjointly. Three of the identified couples were exclusively themes from AI, hinting towards overlapping themes. One couple with both themes created by humans was identified by coder one. Experts agreed on grouping "AI integration and Human-AI collaboration" with "Human-AI Collaboration" for question four, "Common Goals and Objectives" and "Goal Specification" for question two, and the themes "Technology-Enhanced Collaborative Learning Processes" and "Collaborative Learning Process" together. For 87% of the combinations, experts grouped one AI-generated theme and one human-generated theme. Experts estimated the similarity between their picks on average at 3.7/5.

In summary, the automated TA system performed slightly better on the theme creation task, while human codes were preferred for better catching nuance and a better scope of the data. While experts were unable to correctly identify AI-generated themes and codes, the gap is still visible in the data, and heuristics in the distinction between AI and human-coded data indicators could still distinguish outputs.

5 Discussion

The current study examined to what extent an LLM-based MAS can autonomously engage in the iTA process by comparing the output of an automated agent-based system built on Claude 3.5 Sonnet with a manual iTA of four experts. The focus of our analysis is on coding alignment, comparison, and theme development, on a dataset of CSCL survey data. The dataset adheres to criteria of insight novelty, domain specificity, semantic proximity, small data points, and conceptual novelty. To assess the reliability and validity of human-generated and AI-generated codes, we conducted an expert blind review as part of a method to assess both the manual iTA and the outputs of the iTA. Experts provided individual and comparative code assessments before evaluating identified themes along the dimensions of relevance, representativeness, and clarity, before establishing whether themes were close in meaning. Both benchmark dataset criteria and evaluation method

constitute a contribution to RQ1 (What constitutes an appropriate *method* to assess the quality of iTA?).

Regarding RQ2, which examines whether the evaluated system can produce an iTA of similar quality to a human expert team, our results reveal that AI-generated code produces consistent and structured codes closely aligned with human-generated codes. The expert evaluation demonstrates that AI-generated codes were considered appropriate in 91% of cases, surpassing human-generated codes (81%) in individual assessments. This result is consistent with past research [7], emphasizing the ability of AI to enhance coding accuracy while being complementary to human-conducted analysis. At the same time, experts expressed an overall preference for human codes in direct comparison of code sets. Therefore, we conclude that this system, while adequate in generating codes, should be used in a complementary role in human teams, providing a second point of view.

All themes, whether created by AI or the human iTA team, received indistinctively good ratings along the dimensions of clarity, representativeness, and relevance, with one expert favoring AI-created themes for clarity.

Feedback from the human coder team responsible for the human iTA revealed that human coders would revise their theme descriptions to better reflect the essence of these themes based on the AI codebook. This observation aligns with the external expert's preference for AI-generated themes. While the team estimated the automated TA system's output for codes and themes as potential to improve their analysis, experts noted the final report lacked an overall, comprehensive narrative, and was not yet equivalent in quality to the final report of a human iTA.

To the best of our knowledge, the tested system's architecture is novel in the realm of qualitative analysis for the extent to which it autonomously performs successfully multiple steps of iTA collaboratively. This is an important distinction from existing attempts, where AI-only attempts with single LLMs did not yield results comparable to expert iTAs [7] or multi-agent systems with a unidirectional chain-like architecture [25]. Through the results reported in this paper, we observe that, although generative AI systems are closing in on the quality of human iTA expert teams, the method cannot be fully automated yet.

The tendency towards hybrid approaches of humans and generative AI is aligned with literature [8, 17], and this paper demonstrates that hybridity is no longer limited to individual steps of iTA: a trend to AI systems assuming an increasing role in qualitative methods seems to emerge. Given that this tendency towards more autonomy is likely to increase, the urgency for flexible methods to evaluate the quality of not only fully automated systems but also hybrid approaches is real. Based on our results, it seems that code development and theme creation are tasks that can be delegated to MAS, while the final iTA report creation step is still a challenge for these systems.

The study's limitations are related to both the method and the system's novelty. While the system has shown promising performance for automating iTA tasks such as code and theme creation on a domain-specific dataset, further testing on a wider variety of datasets is required to generalize the findings of this study. There is also room to further optimize the existing system through variation of parameters or agent prompts.

6 Conclusion

This study presents one of the first systematic comparisons between a human-led and a fully autonomous multi-agent LLM system (MAS) conducting inductive thematic analysis (iTA). We introduce a novel evaluation framework grounded in expert blind review and benchmark dataset criteria, enabling structured comparison of human and AI-generated qualitative outputs. Our contributions are threefold. First, we propose a method for evaluating automated iTA outputs that integrates code- and theme-level assessments, offering a replicable standard for future studies. Second, we show that a Claude 3.5-based MAS can produce high-quality codes and themes, rated as appropriate in 91% of cases, sometimes even prompting human coders to revise their own outputs. Third, we provide initial evidence that MAS systems can replicate most steps of iTA, (particularly code and theme generation) approaching human-level depth and coherence in analysis. The novelty of this work lies in both the end-to-end automation of iTA mimicking human consensus-building and the introduction of principled evaluation methods. While current systems fall short of generating comprehensive narratives, they offer strong potential as collaborators in qualitative research and, to some extent, can successfully replicate conceptual discussions required to achieve high-quality iTA results. We advocate for hybrid workflows where MAS systems augment human and single-LLM insight, making qualitative analysis more scalable and automated iTA more transparent by documenting intermediate workflow results.

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